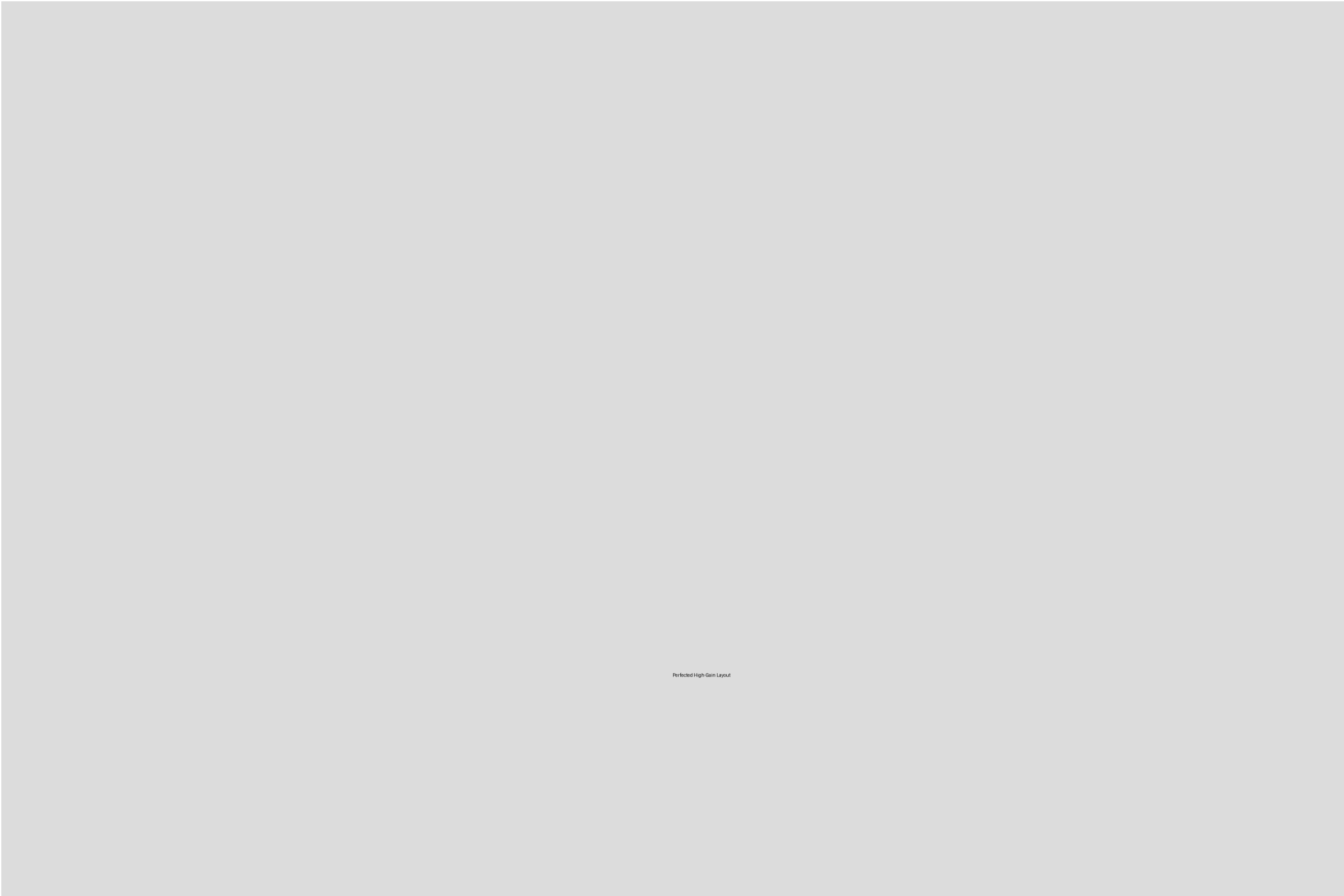


DIY EEG Guide: High-Performance DRL Redesign

High-Gain EEG Schematic (10,000x) with Active DRL

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DESIGN.md

EEG Circuit Design: High-Performance DRL Redesign

Architecture (1 IC / 4 Op-Amps)

1. Precision Instrumentation Amplifier:

- Stage 1 (Op-amps 1, 2): Gain = 101 ($R_g=200\Omega$, $R_1=R_2=10k\Omega$).
- Stage 2 (Op-amp 4): Gain = 100 ($R_{sub}=10k\Omega$, $R_f=1M\Omega$).
- Total System Gain: $\approx 10,100\times$.

2. Frequency Filtering:

- Low-pass: $C_f=4.7nF$ in parallel with $R_f=1M\Omega$ ($f_c \approx 34Hz$).
- High-pass: AC coupling between Electrode buffers.

3. Active Noise Cancellation (DRL - Op-amp 3):

- Inverted common-mode injection via Pin 1 to actively cancel 50Hz hum.

4. Power: Single 9V Battery with 4.5V Virtual Ground.

Bill of Materials

- ICs: 1x TL074CN
- Resistors: 1x 200 (R_g), 11x 10k, 2x 100k (DRL), 3x 1M (R_f , R_{ref} , R_{pd}).
- Capacitors: 1x 4.7n (C_f), 2x 100n (C_c), 1x 47uF (Shunt), 1x 100uF (VGND).

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ASSEMBLY.md

Assembly Guide: High-Gain DRL EEG

Physical Rules

- Battery: Directly under the board.
- IC: Center.
- Parts: Corners (Away from IC).
- Connections: All jumper wires route around the IC body.

Connection Summary

- Electrode A: Pin 12.
- Electrode B: Pin 5.
- Driven Ref (Ear): Pin 1.
- Scope Output: Pin 8 (Use AC Coupling).

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TESTING.md

Safety & Human Capture

Safety

- Battery only is mandatory.
- Scope ground to battery negative.

Alpha Waves

Capture 8-13Hz oscillations by closing eyes while Electrode A/B are on the forehead.

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CONNECTIONS.md

Verified Exhaustive Connection List (10,000x Gain)

1. Power & Virtual Ground (MANDATORY)

- TL074 Pin 4 (VCC+) -> Breadboard Red Rail (+9V)
- TL074 Pin 11 (VCC-) -> Breadboard Blue Rail (0V)
- Rvg1 (10k): Red Rail to VGND Rail
- Rvg2 (10k): VGND Rail to Blue Rail
- Cvg (100uF): VGND Rail (+) to Blue Rail (-)

2. Stage 1: Buffered IA (U1A, U1B)

- Electrode A -> Pin 12 (In+)
- Electrode B -> Pin 5 (In+)
- Rg (200 Ohm): Pin 13 to Pin 6
- R1 (10k): Pin 14 to Pin 13
- R2 (10k): Pin 7 to Pin 6

3. Stage 2: Balanced Difference Amp (U1D)

- Rsub1 (10k): Pin 14 to Pin 9
- Rsub2 (10k): Pin 7 to Pin 10
- Rref (1M): Pin 10 to VGND Rail (Matches Rf for balance)
- Rf (1M): Pin 9 to Pin 8 (Feedback)
- Cf (4.7nF): Pin 9 to Pin 8 (Low-pass filter, $f_c \sim 34\text{Hz}$)

4. Stage 3: Active DRL (U1C)

- Rcm1 (100k): Pin 14 to Pin 2

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- Rcm2 (100k): Pin 7 to Pin 2
- Rdrl (1M): Pin 2 to Pin 1
- Pin 3 (In+) -> VGND Rail
- Pin 1 (DRL Out) -> Driven Reference Electrode (Ear)

5. Scope Output

- Pin 8 (Final signal) -> Scope Probe (Use AC Coupling)
- Blue Rail (0V) -> Scope Ground